EARTHPRESS · DESIGN DECISION RECORD

EarthPress decisions of 2026-10-30

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DOCUMENT CONTROL

Document ID	EPR-DDR-005	Version	0.1
Status	Draft	Date	2026-10-30
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/earthpress		
Source	docs/decisions/0005-decisions-2026-10-30.md @ 212cbb1		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	Records decision D2 (R7 total mass) as decided by Amish on 2026-10-30; D1 (concept) and the design load question stay open because they had no recommendation

0005: Decisions of 2026-10-30

- **Date:** 2026-10-30
- **Status:** accepted (D2); D1 and the design load question remain proposed, awaiting Amish

Context

After a report listing the proposed calls in four repos, Amish wrote on 2026-10-30: "on the calls - i go with your recommendations". Only a decision that carried a stated recommendation is decided by that sentence.

What was decided

#	Decision	Recommendation taken	Result
D2	R7 total mass: the detailed model gives 203.8 kg against the 200 kg total	Option (a): set the R7 total at 205 kg, counting bushes, nuts, washers and weld metal, and keep the 50 kg piece limit; option (b), lightening by about 10 kg, stays a value-engineering item	PR-007 is met on paper: 203.8 kg against 205 kg, a margin of 1.2 kg; the heaviest piece is 44.5 kg against 50 kg

What was not decided

- **D1, concept selection (EPR-DDR-004).** A scores 186, B 181 and C 168 of 245; B and C are within 10 % of A and the options went to Amish without a pick. His sentence does not choose a concept. `concept_chosen` stays unset and the repo stays on option A in the drawings, BOM and build plan.
- **Design load question** (independent review, EPR-STM-001 S7, S9, S10): whether the abuse case stays at 1,000 N at the grip (173 kN) with a grip stop, or is raised to about 1,500 N (about 260 kN) with larger lugs. No recommendation was given, so it stays open.

Consequences

- The model, the BOM and the drawings do not change: the decision moves a limit, not a part. `cad/fidelity.json` and the option A model are unchanged, so the options picture does not need re-rendering for this decision.
- The mass margin is small. Any growth in steel or hardware above 1.2 kg reopens PR-007; the weighed prototype decides (TRL 4, on hold).
- The 205 kg figure is a limit, not a measurement: the model weighs every part at its material's density.